

PackScan Bharat: Legal Metrology Compliance Checker

5 Unique Features & Real-World Implementations

Executive Summary

PackScan Bharat is designed to automate label inspection and statutory compliance under the Legal Metrology Act, 2009 and Packaged Commodities Rules, 2011. To bridge the gap between academic prototyping and production-grade departmental utility, five advanced features have been identified, benchmarked against proven enterprise and government systems.

Comparative Feature Matrix

No.	Feature Name	Core Summary	Real-Life System	Practical Implementation
1	Automated Statutory Notice & Seizure Memo Generator	Automatically links detected non-compliance to exact statutory sections and produces court-admissible PDF memos.	MoRTH e-Challan / Parivahan System	When automated cameras capture an over-speeding vehicle, the system immediately pulls the relevant Motor Vehicles Act clause, attaches the photo evidence, and generates an official, legally enforceable penalty notice.
2	Unit Sale Price (USP) & Math Discrepancy Engine	Mathematically validates packaging rates ($USP = MRP / Net\ Quantity$) to protect consumer transparency and identify illegal pricing.	Blinkit, Zepto, Swiggy Instamart (Catalog Ingestion)	During SKU onboarding, if a 500 g item lists a ₹100 MRP, the platform automatically calculates ₹0.20/g and verifies it against the supplier's declared unit sale price

No.	Feature Name	Core Summary	Real-Life System	Practical Implementation
				before the listing is approved.
3	Tamper-Proof Chain-of-Custody (GPS & IST Lock)	Secures digital evidence against courtroom tampering using GPS coordinates, Indian Standard Time, and SHA-256 cryptographic hashes.	FSSAI FoSCoS Inspector App	During on-site audits, food safety officers capture inspection photos inside the mobile app, which hard-locks the GPS location, hardware UID, and server time, preventing vendors from claiming the inspection happened elsewhere.
4	Rule-Engine Configuration Matrix (Dynamic Rules Engine)	Decouples compliance logic from application code so legal updates and gazette revisions can be configured directly in the database.	GSTN (Goods & Services Tax Network)	When the GST Council updates tax brackets or product exemptions, the portal updates the rule matrix in the database rather than rewriting application code, allowing the calculation API to run with immediate effect.
5	Multi-Database Cross-Verification (FSSAI & GS1 Lookup)	Cross-checks extracted license numbers and product barcodes against official national databases to detect counterfeit packaging.	GS1 India 'Smart Consumer' App	Scanning a product barcode queries GS1's DataKart national repository to fetch authentic brand ownership and verify active FSSAI registration, immediately flagging fraudulent labels that copy authentic branding.